

DEAKIN UNIVERSITY

RESEARCH PROJECT/RESEARCH PROJECT (ADVANCED)

ONTRACK SUBMISSION

Research Progress Check Point 2

Submitted By:

N/a NIKHIL

s224234862

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Tutor:

Ananda MAITI

Task Outcomes		Supports
TLO1	Demonstrate engagement with the following unit learning outcomes (legacy)	ULO3 Communication of Research Outcomes

April 11, 2026



Deakin University

SIT724 Research Project / SIT746 Research Project (Advanced)
OnTrack Submission

Task 3.2P — Research Progress Checkpoint 2

Student Name: Nikhil Sarwara
Student ID: 224234862
Supervisor: Dr. Ananda Maiti
Project Title: Detection and Simulation of GPS Spoofing in
Mobile Robots Using Sensor Cross-Validation
Unit: SIT724 / SIT746 Research Project (Advanced)
Trimester: Trimester 1, 2026
Period Covered: 13 March 2026 – 11 April 2026
Total Meetings: 5
Submitted: 11 April 2026

Contents

1 Overview	3
2 Continuity from Checkpoint 1	4
3 Weekly Progress Diary	5
3.1 Week 1 — Meeting 1 (13 March 2026)	5
3.1.1 Meeting Details	5
3.1.2 Context and Purpose	5
3.1.3 Progress This Week	5
3.1.4 Challenges This Week	5
3.1.5 Decisions and Agreed Next Steps	6
3.1.6 Evidence	6
3.2 Week 2 — Meeting 2 (19 March 2026)	8
3.2.1 Meeting Details	8
3.2.2 Context and Purpose	8
3.2.3 Progress This Week	8
3.2.4 Challenges This Week	9
3.2.5 Mitigation	9
3.2.6 Decisions and Agreed Next Steps	9
3.2.7 Evidence	9
3.3 Week 3 — Meeting 3 (26 March 2026)	12
3.3.1 Meeting Details	12
3.3.2 Context and Purpose	12
3.3.3 Progress This Week	12
3.3.4 Challenges This Week	13
3.3.5 Decisions and Agreed Next Steps	13
3.3.6 Evidence	13
3.4 Week 4 — Meeting 4 (3 April 2026)	16
3.4.1 Meeting Details	16
3.4.2 Context and Purpose	16
3.4.3 Progress This Week	16
3.4.4 Challenges This Week	17
3.4.5 Decisions and Agreed Next Steps	17
3.4.6 Evidence	17
3.5 Week 5 — Meeting 5 (10 April 2026)	22
3.5.1 Meeting Details	22
3.5.2 Context and Purpose	22
3.5.3 Progress This Week	22
3.5.4 Challenges This Week	22
3.5.5 Decisions and Agreed Next Steps	23

3.5.6 Evidence	23
4 Cumulative Milestone Summary (13 March – 11 April 2026)	28
5 Challenges and Mitigations	29
6 Remaining Work (Checkpoint 2 Onwards)	30
7 Reflection	31

Overview

This document is my Checkpoint 2 submission for Task 3.2P of SIT724/SIT746, covering the period from 13 March 2026 to 11 April 2026. It serves as a diary of weekly research progress, evidence of milestone achievements, problems encountered, and how I addressed them, all discussed at regular supervision meetings with Dr. Ananda Maiti.

My research project — *Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation* — is a direct continuation of my previous trimester work (SIT723/SIT792). The prior trimester established a strong academic and technical foundation, including an extensive literature review, a validated research methodology, and a fully operational PX4–Gazebo simulation environment. This trimester has focused on extending that foundation into advanced GPS spoofing simulation, machine learning-based anomaly detection, and real-time inference pipeline development.

During this checkpoint period, I held five regular supervisor meetings and made substantial technical progress across all planned areas.

Meeting #	Date
1	13 March 2026
2	19 March 2026
3	26 March 2026
4	3 April 2026
5	10 April 2026

Continuity from Checkpoint 1

In my Checkpoint 1 submission (Task 3.1P, submitted 17 March 2026), I outlined the following planned next activities:

- Implementing controlled GPS spoofing injection within the simulation
- Expanding the detection module to support advanced anomaly classification
- Generating datasets for machine learning
- Continuing regular supervisor meetings

I am pleased to confirm that all of these objectives have been achieved during this checkpoint period. Not only did I complete the planned implementation work, but I also went significantly beyond the plan by delivering a fully operational real-time ML inference pipeline, a live Streamlit UI dashboard, and a feedback loop back to the simulation — all ahead of schedule.

Weekly Progress Diary

Week 1 — Meeting 1 (13 March 2026)

Meeting Details

Date: 13 March 2026, 05:01 AM (UTC)
Duration: 22 minutes 28 seconds
Mode: Online (screen sharing)
Attendees: Nikhil Sarwara, Dr. Ananda Maiti

Context and Purpose

This was the first meeting of Trimester 1, 2026. I used this meeting to review my previous trimester's work with Dr. Maiti, present my trimester plan, share relevant research papers, and confirm the technical direction for the project going forward.

Progress This Week

- I confirmed all previous trimester work (SIT723/SIT792) had been reviewed, including the literature review, research methodology, and working PX4–Gazebo simulation environment.
- I presented my trimester plan, which included: completing the Docker-based simulation environment, implementing a GPS spoofing simulation module, and developing both attack (spoofing injection) and defence (detection) components.
- I raised a key research direction question with Dr. Maiti: whether to use a traditional algorithmic approach (IMU/sensor fusion, IMAMUL-based) or a machine learning approach (training a model on telemetry data to detect GPS spoofing anomalies). I presented relevant research papers, including an LSTM Transformer for GNSS spoofing and a multi-sensor fusion paper.
- Dr. Maiti guided me to proceed with the **machine learning approach as the primary mechanism**, cross-validated with IMU/sensor data (IMAMUL). This confirmed my research direction for the trimester.
- I informed Dr. Maiti that the Docker simulation environment was ready from the previous trimester and planned to share the Docker image link by Monday, 16 March 2026.

Challenges This Week

- Home internet connectivity was unreliable (NBN connection pending), which prevented a live demo during the meeting. Dr. Maiti suggested using the university's Wi-Fi for future demos.

- I acknowledged limited prior experience with machine learning; however, I expressed confidence in building Python-based models.

Decisions and Agreed Next Steps

- Proceed with ML-based anomaly detection cross-validated with IMU/sensor fusion (IMAMUL).
- Continue exploring and understanding the identified research papers.
- Share Docker image link with Dr. Maiti by Monday, 16 March 2026.
- Focus next meeting on a live demonstration of the Docker simulation environment.

Evidence

Evidence 1.1 — Supervisor Meeting 1 Recording (SIT-723-2)

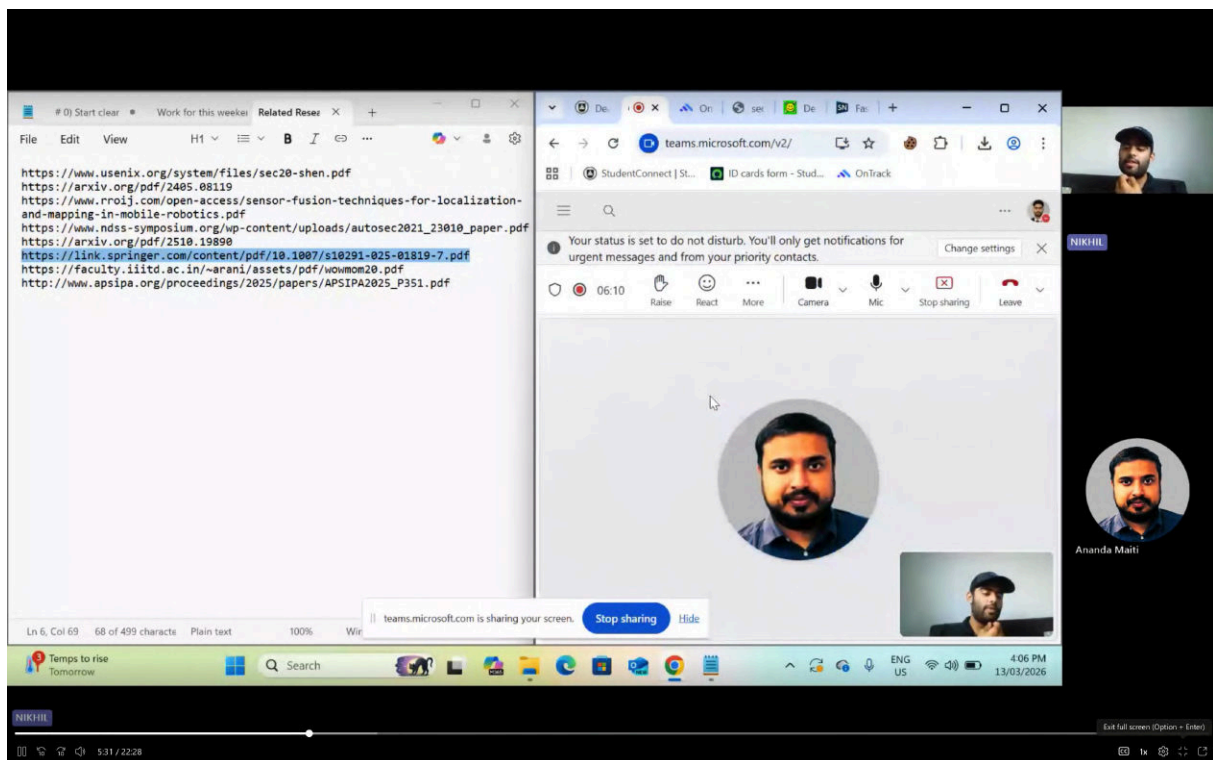


Figure 1: Meeting 1 recording proof — 13 March 2026 (SIT-723-2, 22 min 28 sec)

Evidence 1.2 — Research Papers Reviewed (LSTM / Neural Network GPS Spoofing Literature)

A LSTM Neural Network applied to Mobile Robots Path Planning

Fiorato Nicola, Yasutaka Fujimoto
Department of Electrical and Computer Engineering
Yokohama National University
 Yokohama 240-8501 Japan
 fiorato.nicola@gmail.com, fujimoto@ynu.ac.jp

Roberto Oboe
Department of Engineering and Management
University of Padova
 Vicenza 36100 Italy
 roberto.oboe@unipd.it

Abstract—Mobile robots path planning is a central problem in every situation where human intervention is not desired or not possible to accept: full automated industrial warehouses or general stocking areas and every domestic application that involves a mobile robot and special cases where environment is prohibited for human accessing like toxic wastes and bombs defusing [1]. Currently, neural networks are applied to problems related to mobile robot navigation. However, they are not as popular as in applications like image processing, speech recognition or machine translation, where they are commercially relevant. In this paper we propose a *Long Short-Term Memory* (LSTM) neural network as an online search agent to tackle the problem of mobile robots path planning in unknown environments, meaning that the agent relies only on local map awareness realized with a LRF sensor and relative informations between robot and goal position. Specifically, a final structure of LSTM network is analyzed and its performance is compared with the A* algorithm, a widely known method that follows the best-first search approach. Subsequently, an analysis of the method developed on a real robot is described.

Index Terms—Artificial Intelligence - LSTM - Path-Planning - Mobile Robots

I. INTRODUCTION

Recently, neural networks have been successfully applied in path planning for mobile robots [2] [3] [4] [5]. The prospect of using recurrent neural networks increased in the last few years, since they showed to obtain the highest performances

learning approach where the ground truth is provided by the A* algorithm [9]. Datasets are realized with the path solutions found by the supervisor, then the neural network is trained and eventually it is deployed as a standalone agent.

Supervised learning is faster and easier to perform with respect to reinforcement learning. However, it presents an important limit: the experienced state-action pairs during training are limited to the ones decided by the supervisor, so it leads to a more probable wrong decision during effective deployment when the neural network agent gets itself in a state not experienced in the training process.

Understanding the theoretical concept of planning a path is essential to get the reasons why a neural network can be applied. Starting from the decision theory basics, the agent, given a bunch of input data of different nature, produces preferences as its outputs. So, it creates a relationship between data describing its current environment awareness, easily called *state*, and selects the next *action* in order to take itself from the current state to next state supposed to be in a better position with respect to the goal point. Considering a single pair state and associated action not independent to their neighbours in terms of timesteps, instead the association can be examined in groups of state to a group of actions. Therefore, path planning becomes a variable sequence to sequence problem, the ideal target of recurrent neural networks as it can be addressed also to the tasks mentioned at the beginning of this section.

Figure 2: Research papers reviewed for GPS spoofing detection direction — LSTM Transformer and multi-sensor fusion literature

Week 2 — Meeting 2 (19 March 2026)

Meeting Details

Date: 19 March 2026, 10:06 PM (UTC)
Duration: 35 minutes 37 seconds
Mode: Online (screen sharing)
Attendees: Nikhil Sarwara, Dr. Ananda Maiti

Context and Purpose

The purpose of this meeting was to demonstrate the Docker simulation environment, discuss the critical technical obstacle I had encountered, and confirm the revised architecture for moving forward.

Progress This Week

- I developed a custom Python-based tool called “**Inspector**” — a software utility that allowed me to manage Docker containers (start, stop, monitor) through a simple UI, removing the need for direct terminal management. I demonstrated this to Dr. Maiti during the meeting.
- I had reinstated all components from the previous trimester: Docker containers, PX4 SITL simulation, and QGroundControl.
- I identified a critical technical issue: the **PX4 upstream repository had undergone significant changes**. Previously, PX4 SITL used jMAVSim (a lightweight Java-based simulator) as the default. The PX4 team had changed to a script-based system and removed jMAVSim as the default, which caused all of my previously working scripts to break. Additionally, the new system only provided read access to PX4 logs, not full terminal read/write access inside the Docker container, preventing direct MAVLink connection to QGroundControl.
- To overcome this, I proposed and began implementing a **MAVLink UDP Bridge** — instead of accessing the simulation terminal directly, data would be streamed out via a UDP port on the host machine, where my Python tooling could read and write to it. This became the foundation of the entire detection pipeline architecture.
- I migrated the primary simulation and development environment to my **MacBook M1** for significantly better performance. This became the main development platform going forward.
- Dr. Maiti reinforced that the project goal is to *detect* GPS spoofing and propose a credible method backed by simulation results. He also emphasised that writing quality and research presentation are critical for grades.

- Dr. Maiti outlined upcoming deadlines: Literature Review and Research Methodology (Tasks 2.1 and 2.2) by end of April; Research Presentation Recording (Task 4.3D) due approximately 13 April 2026.
- Dr. Maiti recommended using **SciSpace** (an AI-assisted academic writing tool) for the literature review, with targeted references as inputs.

Challenges This Week

- The PX4 upstream change broke all existing scripts and removed jMAVSim as the default — this was the most critical technical obstacle of the trimester so far.
- No read/write terminal access was available inside the Docker container under the new script-based PX4 system.
- MAVLink connectivity to QGroundControl was temporarily non-functional.

Mitigation

I proposed and implemented a MAVLink UDP Bridge to route data from the simulation to an external Python process, bypassing the access restrictions entirely. This architectural shift resolved the blocking issue and formed the basis for all subsequent pipeline work.

Decisions and Agreed Next Steps

- Resolve MAVLink connectivity using the UDP bridge approach.
- Use MacBook M1 as the primary simulation and development machine.
- Begin logging GPS, IMU, and altimetry data via MAVLink once connectivity is restored.
- Begin writing literature review and methodology around 20 April 2026.
- Prioritise the research presentation recording (Task 4.3D, due approx. 13 April 2026).

Evidence

Evidence 2.1 — Supervisor Meeting 2 Recording (SIT-723-3)

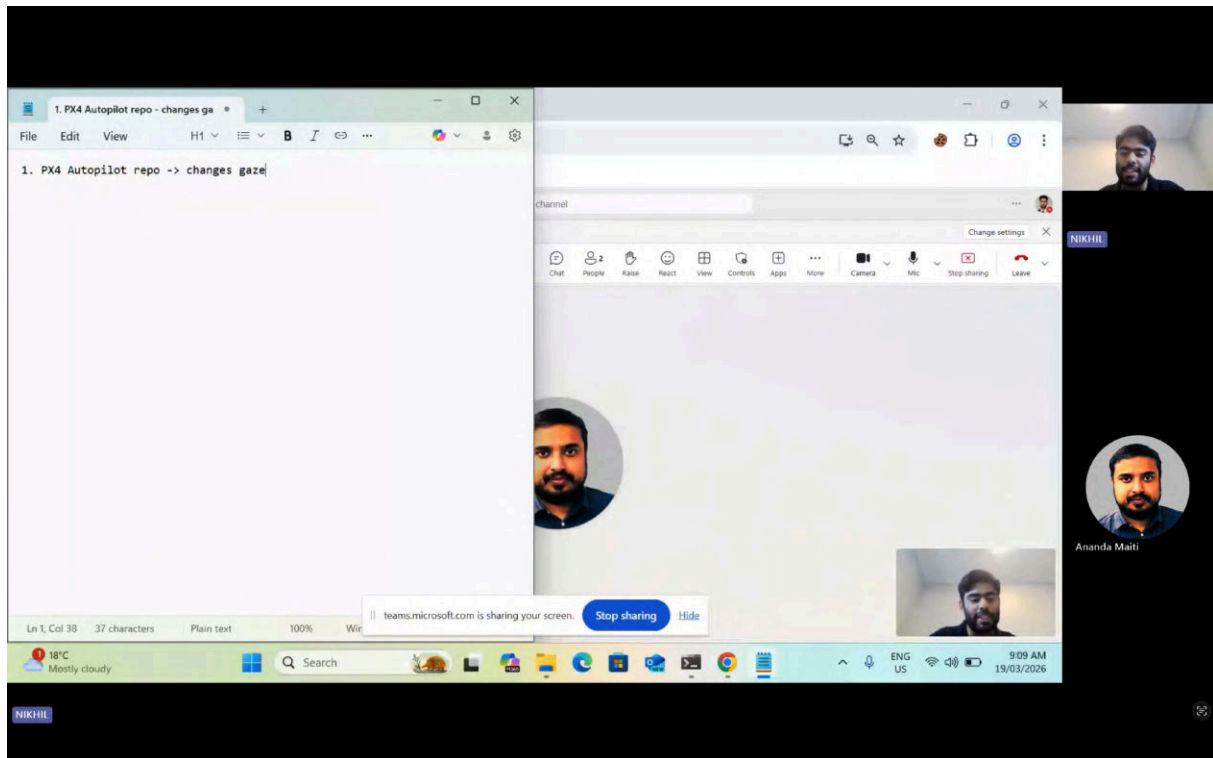


Figure 3: Meeting 2 recording proof — 19 March 2026 (SIT-723-3, 35 min 37 sec)

Evidence 2.2 — Inspector Tool Built in Week 2

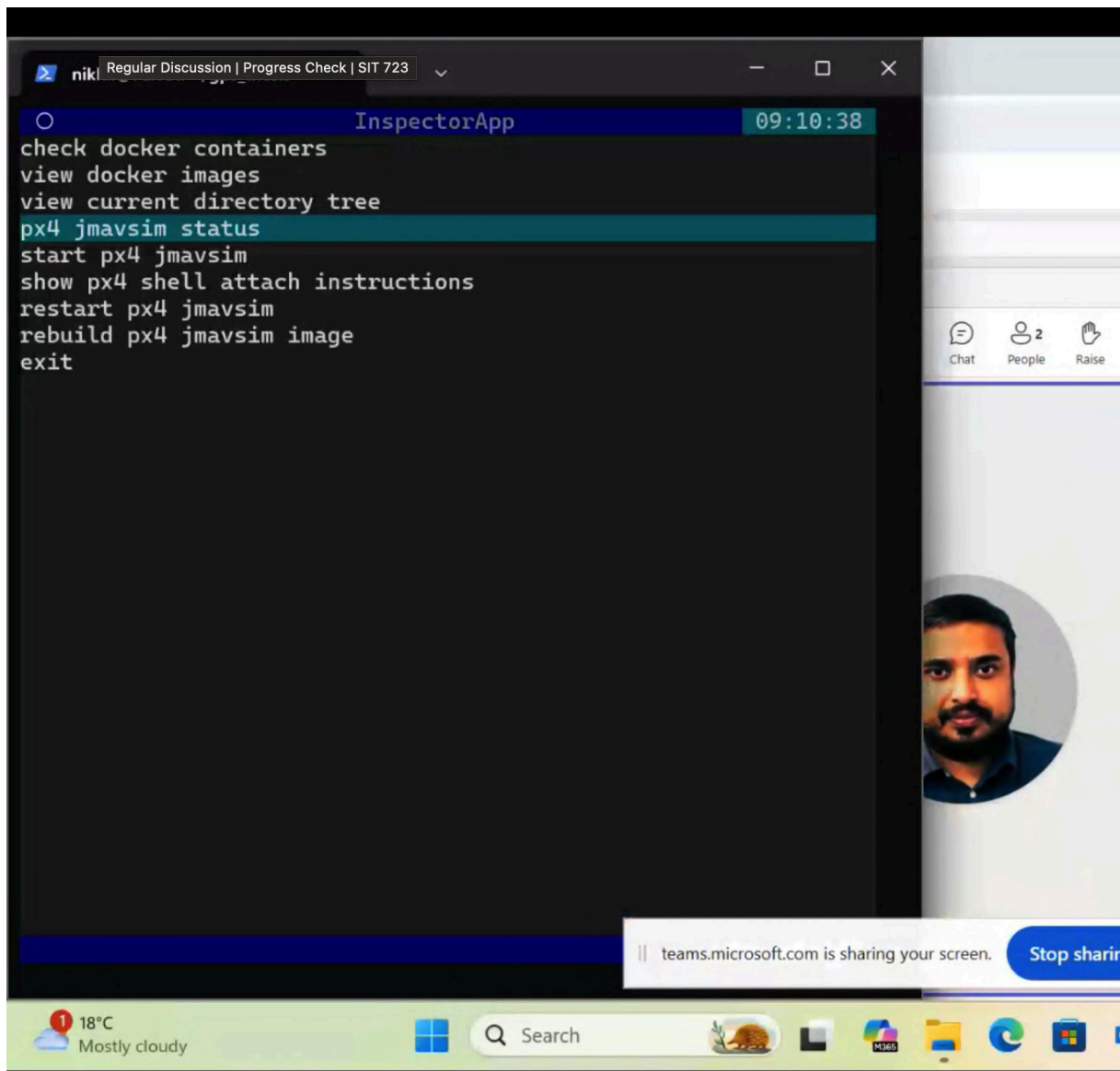


Figure 4: Custom Python Inspector tool for Docker container management — developed and demonstrated in Week 2

Week 3 — Meeting 3 (26 March 2026)

Meeting Details

Date: 26 March 2026, 10:00 PM (UTC)
Duration: 35 minutes 39 seconds
Mode: Online (screen sharing)
Attendees: Nikhil Sarwara, Dr. Ananda Maiti

Context and Purpose

The purpose of this meeting was to demonstrate the working telemetry logging pipeline, discuss the ML architecture design, clarify how the ML model would interface with the simulation, and agree on next steps for model development.

Progress This Week

- **Simulation environment fully migrated to MacBook M1:** The complete Docker-based PX4–Gazebo simulation environment was running reliably on my MacBook, with the MAVLink UDP Bridge fully operational.
- **Telemetry logging pipeline operational:** I successfully implemented data capture of GPS logs, IMU (IMAMUL) data, and altimetry data via MAVLink. The data pipeline logged: timestep, altitude, speed, GPS location (latitude, longitude), heading, velocity, and multiple other telemetry parameters.
- **Dataset creation:** I set up sliding window datasets, processing telemetry data into fixed-size windows and labelling them as normal or anomalous (spoofed) behaviour.
- **Drone control confirmed:** I confirmed I could now also actively control the drone within the simulation, enabling more varied flight path generation.
- **ML baseline infrastructure complete:** All prerequisite infrastructure for building the machine learning model was in place.
- I demonstrated the flight logging output live to Dr. Maiti during the meeting.
- I proposed hosting the ML model as an **external Python process** on the host machine, receiving live telemetry from the simulation via a UDP MAVLink port. Dr. Maiti confirmed this architecture was perfectly acceptable.
- Dr. Maiti gave strong positive feedback: “*You are doing way better than anybody else I have seen*” and “*You are definitely on track for HD.*”
- Dr. Maiti instructed me to prepare a **system flowchart / algorithm diagram** showing exactly how ML will be used and present it at the next meeting.

Challenges This Week

- Limited experience with MAVLink SDK and plugin development — resolved by confirming that the external process via UDP is a valid research approach.
- Initial uncertainty about how to embed the ML model inside the simulation — resolved by Dr. Maiti's architectural guidance.

Decisions and Agreed Next Steps

- Generate 500–1000 labelled mission datasets from the simulation.
- Train the ML model on an 80/20 train/test split.
- Prepare system flowchart / algorithm diagram — present at next meeting.
- ML model to be hosted as an external Python process via UDP/MAVLink.
- Begin writing from approximately 20 April 2026.

Evidence

Evidence 3.1 — Supervisor Meeting 3 Recording (SIT-723-4)

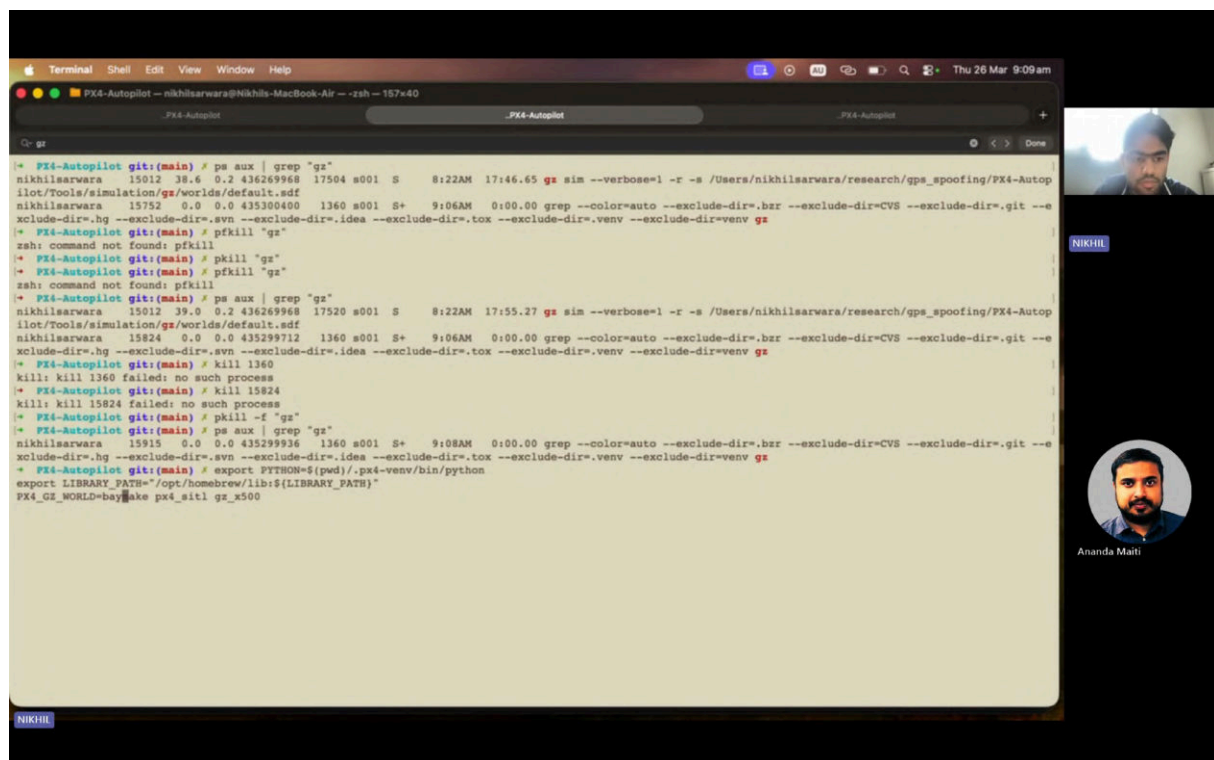


Figure 5: Meeting 3 recording proof — 26 March 2026 (SIT-723-4, 35 min 39 sec)

Evidence 3.2 — Cleaned and Processed JSON Telemetry File Created


```

..ofing/project                                     ..PX4-Autopilot
"input_file": "gps_logs/raw/gps_log_20260402_184048.csv",
"output_file": "gps_logs/processed/gps_log_20260402_184048_cleaned.csv",
"initial_rows": 2783,
"final_rows": 1851,
"rows_removed": 932,
"time_start_s": 0.105,
"time_end_s": 269.557,
"duration_s": 269.452,
"missing_after_cleaning": {
  "time_s": 0,
  "lat_deg": 0,
  "lon_deg": 0,
  "alt_m": 0,
  "rel_alt_m": 0,
  "vel_m_s": 0,
  "hdg_deg": 0,
  "fix_type": 0,
  "satellites_visible": 0,
  "eph_m": 0,
  "epv_m": 0,
  "ax_mps2": 1851,
  "ay_mps2": 1851,
  "az_mps2": 1851,
  "gx_radps": 1851,
  "gy_radps": 1851,
  "gz_radps": 1851,
  "battery_voltage": 2,
  "battery_remaining_pct": 2,
  "armed": 0,
  "mode": 0,
  "failsafe": 0,
  "connection_ok": 0,
  "last_update_iso": 0,
  "is_stale_repeat": 0,
  "dt": 1
},
"mode_counts": {
  "LOITER": 891,
  "MODE_HEX": 780,
  "TAKEOFF": 116,
  "LAND": 62,
  "UNKNOWN": 2
},
"failsafe_counts": {
  "1": 1662,
  "0": 189
},
},

```

Figure 6: Cleaned and processed JSON telemetry file output from the MAVLink data pipeline

Evidence 3.3 — CSV Dataset Inspection

```

..ofing/project                                     ..PX4-Autopilot
(.px4-venv) ➔ project git:(master) ✗ python - <<'PY'
import pandas as pd
df = pd.read_csv("gps_logs/processed/gps_log_20260402_184048_cleaned.csv")
print(df.head())
print(df.tail())
print(df[["time_s", "mode", "armed", "failsafe", "connection_ok"]].head(20))
print(df.shape)
PY
   time_s  lat_deg  lon_deg  alt_m  rel_alt_m  vel_m_s  ...  mode  failsafe  connection_ok  last_update_iso  is_stale_repeat  dt
0    0.105  37.412173 -121.998879  38.161    0.046    0.0  ...  UNKNOWN    0            1  2026-04-02T18:40:48.128764    False    NaN
1    0.206  37.412173 -121.998879  38.161    0.046    0.0  ...  UNKNOWN    0            1  2026-04-02T18:40:48.228501    False    0.101
2    0.306  37.412173 -121.998879  38.160    0.046    0.0  ...  LOITER    0            1  2026-04-02T18:40:48.328295    False    0.100
3    0.407  37.412173 -121.998879  38.160    0.045    0.0  ...  LOITER    0            1  2026-04-02T18:40:48.428326    False    0.101
4    0.509  37.412173 -121.998879  38.159    0.044    0.0  ...  LOITER    0            1  2026-04-02T18:40:48.527984    False    0.102

[5 rows x 26 columns]
   time_s  lat_deg  lon_deg  alt_m  rel_alt_m  vel_m_s  ...  mode  failsafe  connection_ok  last_update_iso  is_stale_repeat  dt
1846  269.146  37.412033 -121.998767  37.835    -0.002    0.01  ...  MODE_HEX    1            1  2026-04-02T18:45:17.171116    False    0.100
1847  269.251  37.412033 -121.998767  37.836    -0.001    0.01  ...  LOITER    0            1  2026-04-02T18:45:17.270494    False    0.105
1848  269.353  37.412033 -121.998767  37.838    0.000    0.02  ...  LOITER    0            1  2026-04-02T18:45:17.370873    False    0.102
1849  269.454  37.412033 -121.998767  37.839    0.001    0.02  ...  LOITER    0            1  2026-04-02T18:45:17.470801    False    0.101
1850  269.557  37.412033 -121.998767  37.841    0.003    0.02  ...  LOITER    0            1  2026-04-02T18:45:17.570905    False    0.103

[5 rows x 26 columns]
   time_s  mode  armed  failsafe  connection_ok
0    0.105  UNKNOWN    0         0              1
1    0.206  UNKNOWN    0         0              1
2    0.306  LOITER    0         0              1
3    0.407  LOITER    0         0              1
4    0.509  LOITER    0         0              1
5    0.614  LOITER    0         0              1
6    0.719  LOITER    0         0              1
7    0.822  LOITER    0         0              1
8    0.926  LOITER    0         0              1
9    1.026  MODE_HEX    1         1              1
10   1.129  MODE_HEX    1         1              1
11   1.231  MODE_HEX    1         1              1
12   1.336  LOITER    0         0              1
13   1.438  LOITER    0         0              1
14   1.538  LOITER    0         0              1
15   1.641  LOITER    0         0              1
16   1.744  LOITER    0         0              1
17   1.845  LOITER    0         0              1
18   1.945  LOITER    0         0              1
19   2.051  MODE_HEX    1         1              1
(1851, 26)
(.px4-venv) ➔ project git:(master) ✗

```

Figure 7: Inspection of the cleaned CSV telemetry dataset showing captured GPS and flight fields

Evidence 3.4 — System Data Flow Diagram

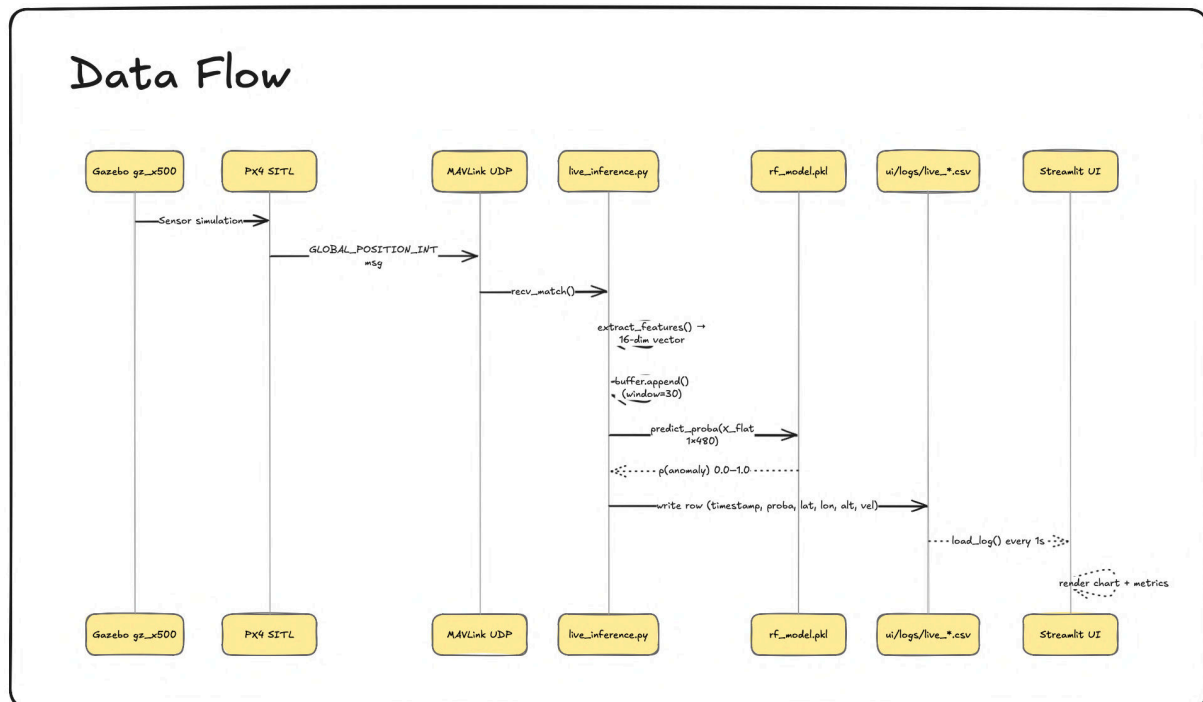


Figure 8: System data flow diagram showing telemetry capture, processing, and ML inference path

Week 4 — Meeting 4 (3 April 2026)

Meeting Details

Date: 3 April 2026
Duration: 37 minutes 20 seconds
Mode: Online (screen sharing)
Attendees: Nikhil Sarwara, Dr. Ananda Maiti

Context and Purpose

The purpose of this meeting was to present the completed baseline machine learning models, demonstrate the GPS Monitor project and live inference pipeline, and discuss the research scope around GPS confidence and spoofing response strategies.

Progress This Week

- **Baseline ML models trained and validated:** I trained and validated two machine learning models — a **Random Forest classifier** and a **Convolutional Neural Network (CNN)**. Both models achieved **100% detection accuracy** on labelled anomalies in the test set.
- **Dataset details:** The labelled dataset consisted of 122+ sliding windows, each 3 seconds long, with 16 features, split 85% training / 15% testing.
- **GPS Monitor project demonstrated:** I demonstrated the live GPS Monitor Python project to Dr. Maiti, showing real-time telemetry being ingested from the simulation.
- **Data labelling and window creation:** I explained the window creation and labelling process, including how spoofing events were injected and captured in the dataset.
- **IMU data access issue:** I raised an issue with accessing IMU (IMAMUL) data — this remained a challenge, as the pipeline currently relied on 16 GPS-based telemetry features only.
- **Live real-time inference pipeline operational:** I built and demonstrated a real-time ML inference pipeline connected to the MAVLink stream using 30-message buffering and stacked window creation.
- **Feedback pipeline built:** I implemented a mechanism to send detection results back to the simulation via MAVLink, completing the detect-and-respond loop.
- **Streamlit UI dashboard:** I built a live monitor and log replay dashboard using Streamlit, demonstrated live to Dr. Maiti.

- Dr. Maiti discussed the concept of a “**breaking point**” — the spoofing intensity at which detection fails — framing this as the key novel contribution of the research.

Challenges This Week

- IMU (IMAMUL) data was not yet integrated into the ML pipeline.
- The anomaly detection threshold (0.5) was fixed and had not yet been formally tuned or justified.

Decisions and Agreed Next Steps

- Integrate IMAMUL data into the ML pipeline.
- Implement the cross-sensor validation algorithm.
- Generate more diverse flight datasets.
- Measure and document the **breaking point** as a core quantitative result.
- Record and submit the video presentation for Task 4.3D.

Evidence

Evidence 4.1 — Supervisor Meeting 4 Recording (SIT-723-5)

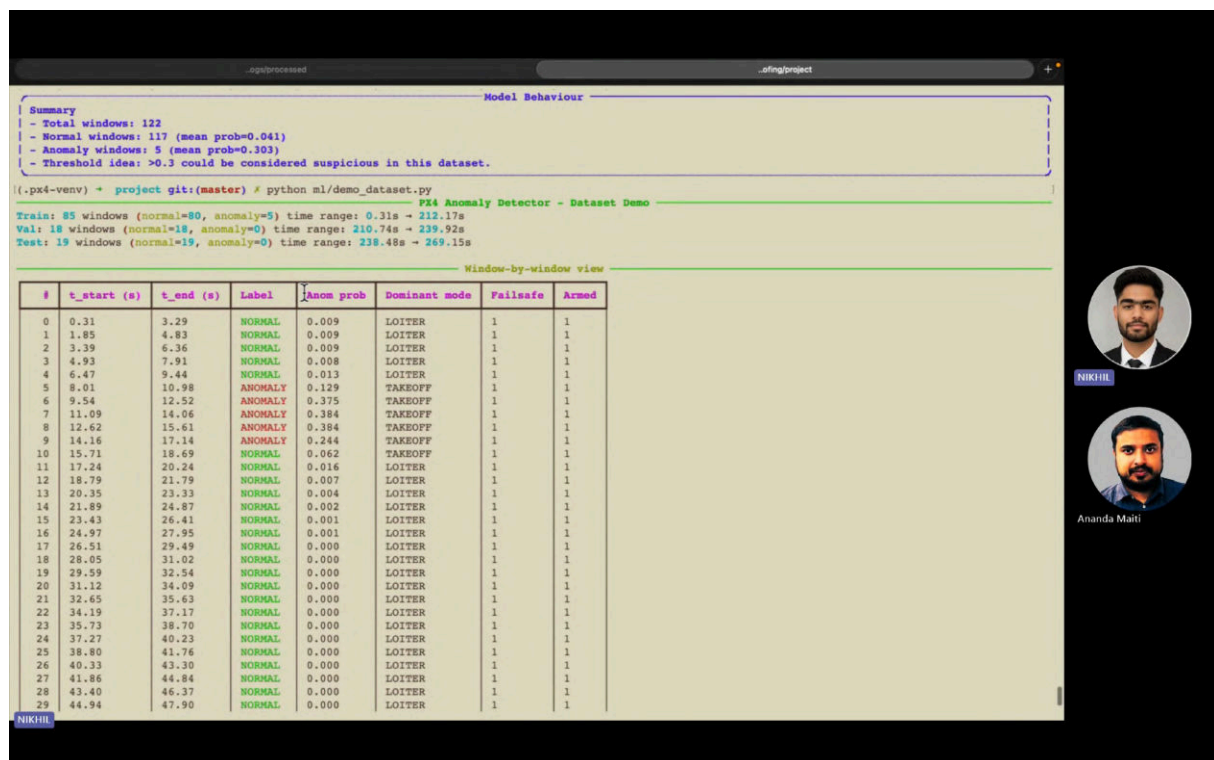


Figure 9: Meeting 4 recording proof — 3 April 2026 (SIT-723-5, 37 min 20 sec)

Evidence 4.2 — Data Label Segments

```
[(.px4-venv) → project git:(master) ✕ python ml/scripts/02_label_segments.py
Segments:
  0.0-2.0s: ignore (startup noise)
  2.0-9.0s: 0 (normal LOITER)
  17.0-166.0s: 0 (normal LOITER)
  170.0-269.0s: 0 (normal LOITER)
  9.0-17.0s: 1 (MODE_HEX/failsafe/stale)

Label distribution:
{0: 1773, 1: 78}

Saved: gps_logs/processed/segments.json
Saved: gps_logs/processed/row_labels.csv

Next: run 03_make_windows.py
(.px4-venv) → project git:(master) ✕ █
```

Figure 10: Label segment annotations applied to telemetry data distinguishing normal vs. spoofed windows

Evidence 4.3 — Labels vs. Time Plot

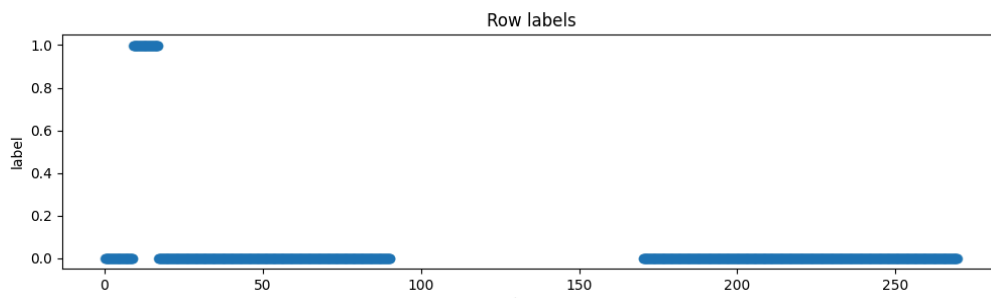


Figure 11: Labels plotted against time showing the distribution of normal and anomalous flight segments

Evidence 4.4 — Sliding Window Creation for Training, Validation, and Testing

```
[(.px4-venv) → project git:(master) ✕ python ml/scripts/03_make_windows.py
Loading data...
Cleaned shape: (1851, 26)
Labeled rows: 1851, anomaly ratio: 4.2%
Converting GPS...
After trim: 1849 rows
Creating windows...
Windows created: (122, 30, 16)
Label dist: [117  5]

✅ Dataset ready in ml/artifacts/
Train: (85, 30, 16), labels={0: 80, 1: 5}
Val:   (18, 30, 16), labels={0: 18}
Test:  (19, 30, 16), labels={0: 19}
Saved:
- ml/artifacts/dataset.npz
- ml/artifacts/scaler.pkl
- ml/artifacts/feature_names.json
- ml/artifacts/dataset_info.json
(.px4-venv) → project git:(master) ✕ █
```

Figure 12: Sliding window generation pipeline for creating train, validation, and test splits from labelled telemetry data

Evidence 4.5 — Label Segment Verification

```
(.px4-venv) → project git:(master) X cat gps_logs/processed/segments.json | jq .
[head gps_logs/processed/row_labels.csv
[
  {
    "start_s": 0.0,
    "end_s": 2.0,
    "label": "ignore",
    "reason": "startup noise"
  },
  {
    "start_s": 2.0,
    "end_s": 9.0,
    "label": 0,
    "reason": "normal LOITER"
  },
  {
    "start_s": 17.0,
    "end_s": 166.0,
    "label": 0,
    "reason": "normal LOITER"
  },
  {
    "start_s": 170.0,
    "end_s": 269.0,
    "label": 0,
    "reason": "normal LOITER"
  },
  {
    "start_s": 9.0,
    "end_s": 17.0,
    "label": 1,
    "reason": "MODE_HEX/failsafe/stale"
  }
]
time_s,label,mode,armed,failsafe,is_stale_repeat
0.105,0,UNKNOWN,0,0,False
0.206,0,UNKNOWN,0,0,False
0.306,0,LOITER,0,0,False
0.407,0,LOITER,0,0,False
0.509,0,LOITER,0,0,False
0.614,0,LOITER,0,0,False
0.719,0,LOITER,0,0,False
0.822,0,LOITER,0,0,False
0.926,0,LOITER,0,0,False
(.px4-venv) → project git:(master) X
```

Figure 13: Verification output confirming that label segments were correctly applied to the dataset

Evidence 4.6 — Streamlit UI Running Live Detection

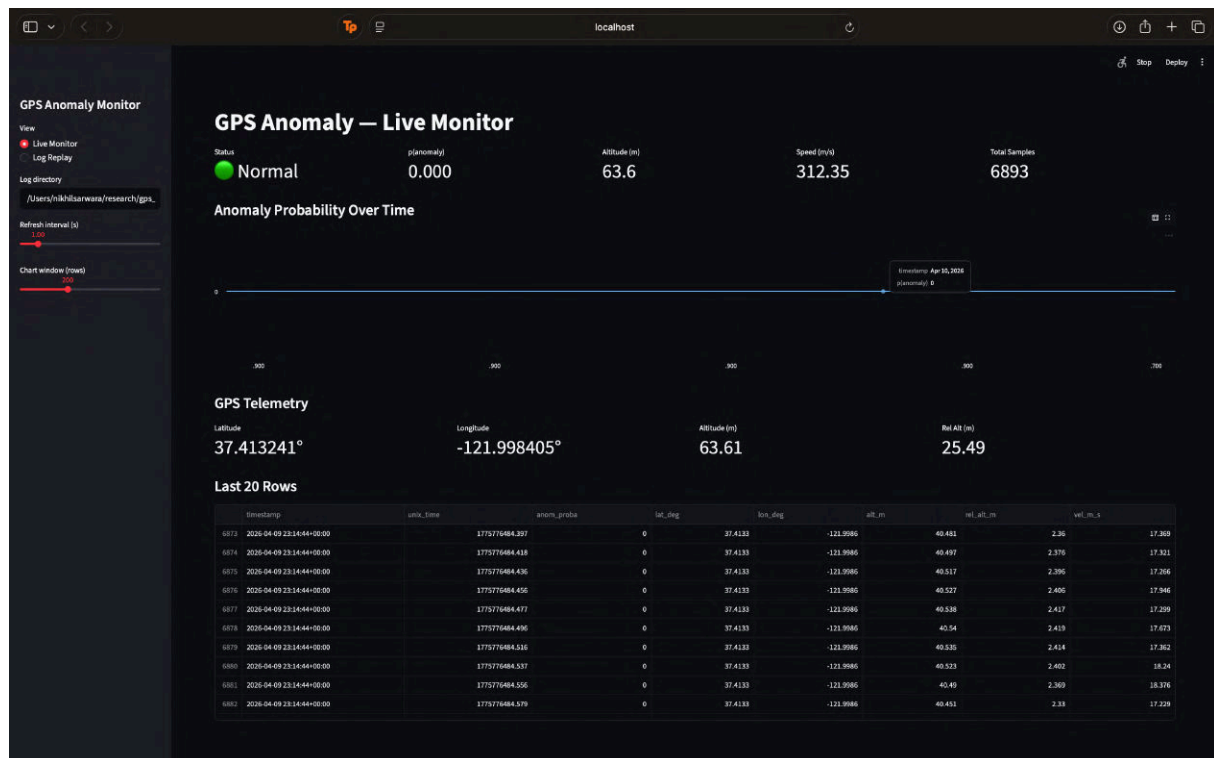


Figure 14: Streamlit live detection dashboard running in real time, showing GPS anomaly detection results from the ML inference pipeline

Week 5 — Meeting 5 (10 April 2026)

Meeting Details

Date: 10 April 2026
Duration: 36 minutes 56 seconds
Mode: Online (screen sharing)
Attendees: Nikhil Sarwara, Dr. Ananda Maiti

Context and Purpose

The purpose of this meeting was to present the full integrated system, explain the complete architecture, receive feedback on the ML pipeline, and receive guidance for the upcoming video presentation (Task 4.3D).

Progress This Week

- **Full system architecture demonstrated:** I explained and demonstrated the complete end-to-end architecture to Dr. Maiti: PX4 SITL in Docker, MAVLink UDP streaming, GPS Monitor project, sliding window creation, ML inference, Streamlit UI, and the feedback loop back to the simulation.
- **System flowcharts completed:** I produced full architecture diagrams covering the system data flow, ML inference pipeline, and Streamlit UI layout.
- **Supervisor feedback on ML pipeline:** Dr. Maiti confirmed the approach was sound and the current state of the system was impressive.
- **Key research contribution confirmed — Breaking Point and Quantitative Analysis:** Dr. Maiti confirmed that the novel contribution is the **quantitative characterisation of GPS spoofing detection** — specifically, at what spoofing intensity detection begins to degrade (the breaking point).
- **Video Presentation Guidance (Task 4.3D, due approx. 13 April 2026):** Dr. Maiti guided me to include: simulation demo, new ML approach, result goals, real-time lag, pipeline overview, and planned result metrics.
- **Publication potential discussed:** Dr. Maiti noted that with strong results, a high-quality journal paper is achievable.

Challenges This Week

- IMU (IMAMUL) data still not fully integrated — pipeline relies on 16 GPS-based features.
- Real-time lag not yet formally quantified.
- Anomaly detection threshold (0.5) still requires tuning.

- Cross-sensor validation algorithm not yet implemented.

Decisions and Agreed Next Steps

- Record and submit the video presentation for Task 4.3D by approximately 13 April 2026.
- Focus on algorithm development (cross-sensor validation) at the next meeting.
- Measure and report the breaking point as a core result metric.
- Log more diverse flight data (varied paths) to improve model robustness.
- Continue IMAMUL integration into the pipeline.
- Begin planning the paper structure and contribution statement.

Evidence

Evidence 5.1 — Supervisor Meeting 5 Recording (SIT-723-6)

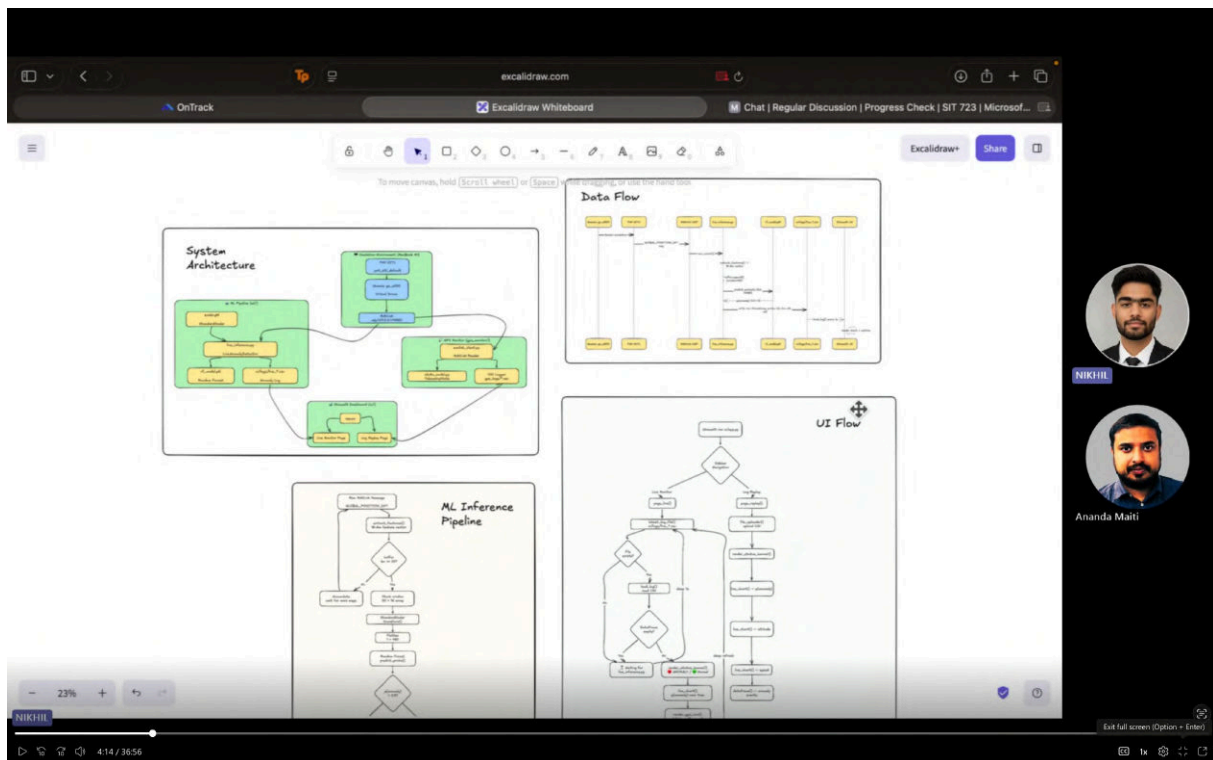


Figure 15: Meeting 5 recording proof — 10 April 2026 (SIT-723-6, 36 min 56 sec)

Evidence 5.2 — System Architecture Diagram

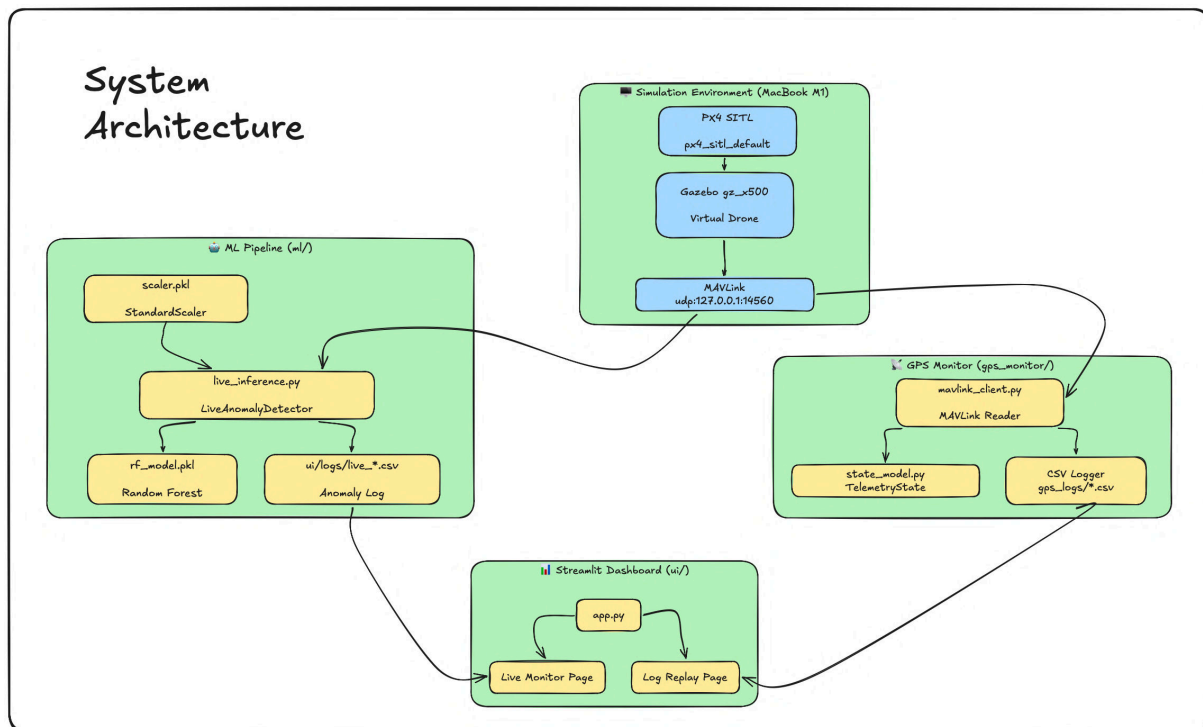


Figure 16: Complete system architecture diagram showing PX4 SITL, MAVLink bridge, ML inference, and feedback loop

Evidence 5.3 — ML Inference Pipeline Diagram

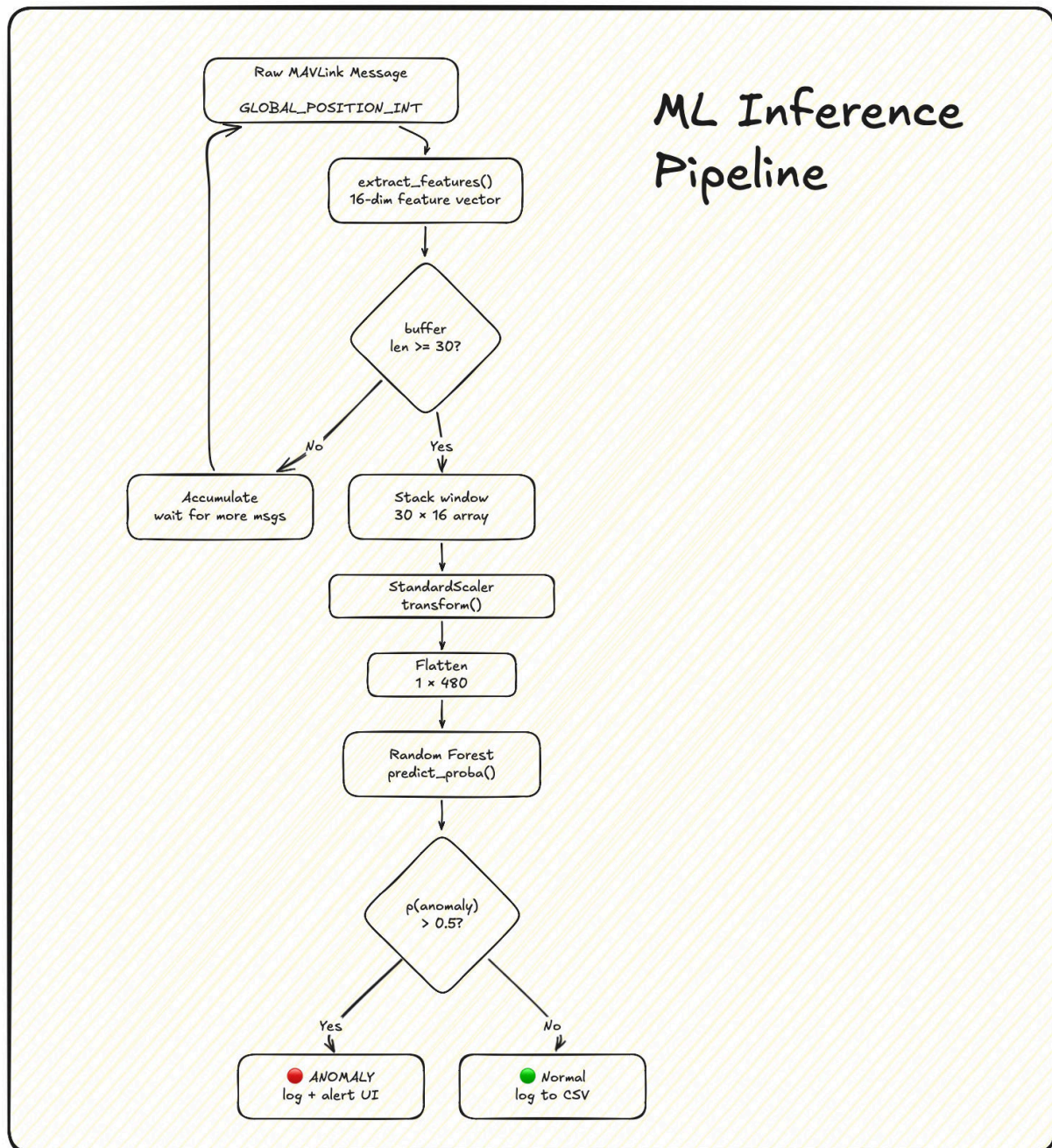


Figure 17: ML inference pipeline diagram: telemetry ingestion, sliding window creation, model prediction, and anomaly flagging

Evidence 5.4 — UI Flow Diagram

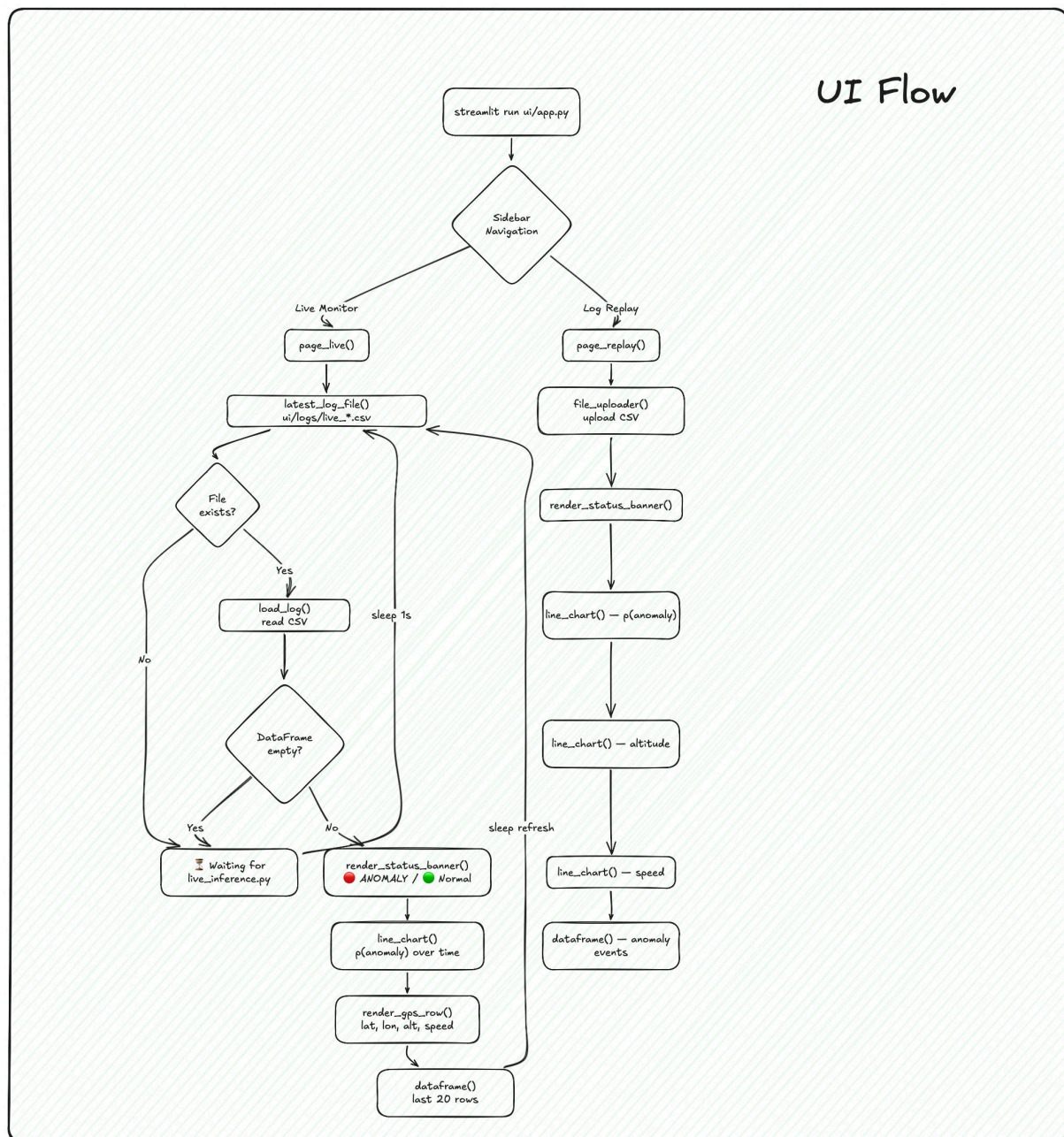


Figure 18: Streamlit UI flow diagram illustrating the live monitoring and log replay interface layout

Evidence 5.5 — Research Project Timeline Graph

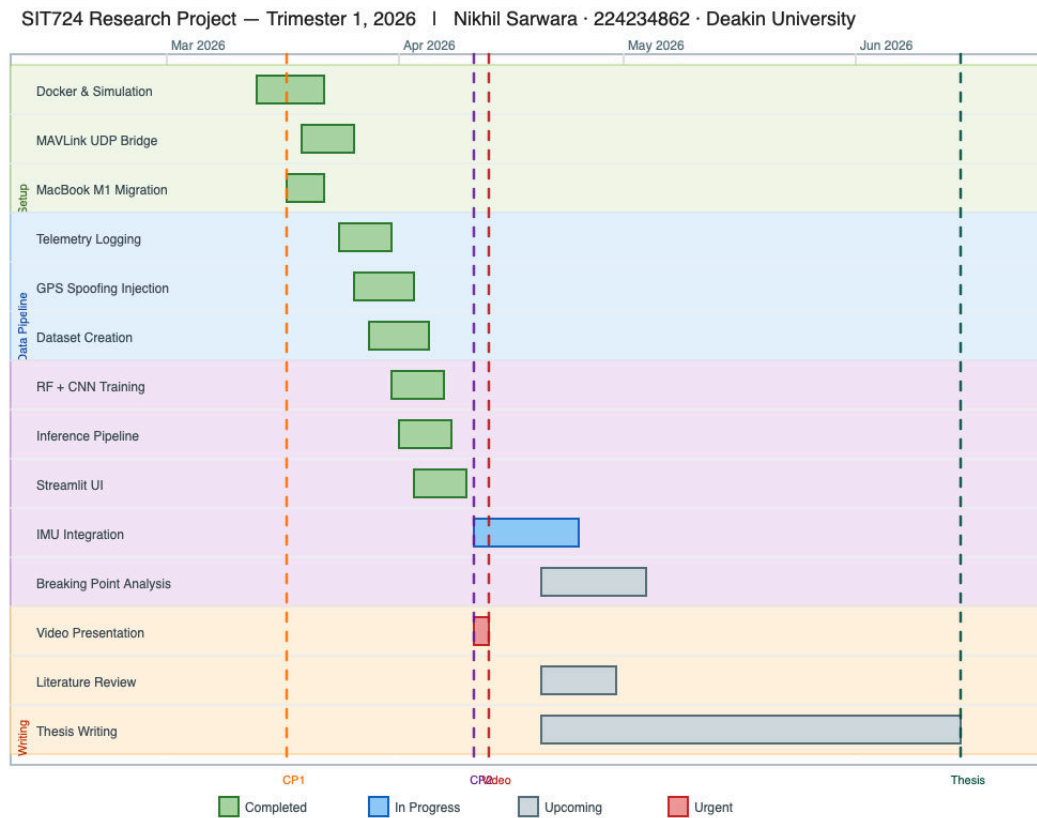


Figure 19: Research project Gantt chart covering Trimester 1, 2026 — all phases, tasks, and milestones

Cumulative Milestone Summary (13 March – 11 April 2026)

The following milestones were achieved during this checkpoint period:

1. **Simulation Environment:** PX4 SITL + Gazebo running in Docker on MacBook M1, with MAVLink UDP connectivity fully operational.
2. **Inspector Tool:** Custom Python-based Docker management UI built and demonstrated.
3. **MAVLink UDP Bridge:** Implemented to resolve PX4 upstream access restrictions and restore full data pipeline connectivity.
4. **Telemetry Logging Pipeline:** GPS logs, altimetry, velocity, heading, and multiple telemetry fields successfully captured via MAVLink in real time.
5. **Dataset Generation:** Labelled sliding-window dataset created — 122+ windows, 3-second windows each, 16 features per window, 85% training / 15% testing split.
6. **GPS Spoofing Injection:** Controlled GPS spoofing scenarios implemented in the simulation and captured in the dataset.
7. **Baseline ML Models:** Random Forest and CNN models trained, validated, and achieving **100% anomaly detection accuracy** on the labelled test set.
8. **Real-Time Inference Pipeline:** Live MAVLink-connected ML inference pipeline operational with 30-message buffering and stacked window creation.
9. **Feedback Pipeline:** Mechanism in place to send detection results back to the simulation via MAVLink, completing the closed detect-and-respond loop.
10. **Streamlit Dashboard:** Live monitor and log replay UI built and demonstrated.
11. **System Flowcharts:** Architecture, data flow, ML inference pipeline, and UI diagrams completed.
12. **Supervisor Meetings:** Five regular meetings held, each with documented discussions, feedback, and agreed actions.

Challenges and Mitigations

Challenge	Mitigation
PX4 upstream repository change removed jMAVSim as default simulator; all existing scripts broke and container access was restricted to read-only.	Designed and implemented a MAVLink UDP Bridge to stream data from the container to an external Python process on the host, bypassing access restrictions entirely.
No read/write terminal access inside the Docker container under the new script-based PX4 system.	Confirmed with supervisor that hosting the ML model as an external Python process receiving data via UDP is architecturally valid for the research.
Home internet connectivity unreliable (NBN pending) during early weeks. IMU (IMAMUL) data not yet fully integrated into the ML pipeline.	Used university Wi-Fi for live demonstrations during supervision meetings. Identified as a planned next step; current pipeline uses 16 GPS-based telemetry features. IMAMUL integration is targeted for the next checkpoint period.
Real-time lag not yet formally quantified.	Flagged as a result metric to be measured and documented as part of the breaking point analysis.
Anomaly detection threshold (0.5) is fixed and not yet formally tuned. Cross-sensor validation algorithm not yet implemented.	Flagged for tuning and justification as part of upcoming quantitative analysis work. Baseline statistical detection is currently in place. Full cross-sensor validation is the primary remaining algorithmic task.

Remaining Work (Checkpoint 2 Onwards)

The following work remains to be completed from this point:

1. Integrate IMAMUL (IMU) data into the ML pipeline for full cross-sensor validation.
2. Implement the cross-sensor validation algorithm.
3. Generate more diverse and varied flight datasets to improve model robustness.
4. Measure, characterise, and document the **breaking point** — the spoofing intensity at which detection performance begins to degrade — as the core quantitative result and novel contribution.
5. Formally quantify real-time pipeline lag.
6. Tune and justify the anomaly detection threshold.
7. Record and submit the **video presentation (Task 4.3D)**, due approximately 13 April 2026.
8. Write the **Literature Review and Research Methodology (Tasks 2.1 and 2.2)** by end of April 2026.
9. Begin thesis writing from approximately 20 April 2026.
10. Shift primary focus to thesis writing from mid-May 2026.
11. Plan paper structure and contribution statement in consultation with Dr. Maiti.

Reflection

Reflecting on this checkpoint period, I am satisfied with the substantial progress made. What began as a difficult week — encountering a critical upstream PX4 repository change that broke all my existing infrastructure — ultimately led to a more robust and architecturally sound system. The MAVLink UDP Bridge I designed to overcome that obstacle became the backbone of the entire detection pipeline.

By the end of this period, I had gone well beyond my Checkpoint 1 plan. I had not only completed the spoofing injection and dataset generation as planned, but had also trained functional ML models, built a real-time inference pipeline, implemented a feedback loop to the simulation, and developed a Streamlit UI for live monitoring and presentation. Dr. Maiti's feedback — that I was on track for an HD and performing well above average — was encouraging and reinforced my confidence in the direction taken.

The most important insight from these five meetings was the clarity around my novel research contribution: the **quantitative characterisation of the breaking point** — when and how GPS spoofing detection degrades as attack intensity increases. This is what will form the scientific core of my thesis and will distinguish this work from existing literature.

Going forward, my focus will be on integrating IMU cross-validation, generating the breaking point data, and transitioning into thesis writing from 20 April 2026.

Submitted by Nikhil Sarwara (224234862) — SIT724/SIT746 Research Project, Trimester 1, 2026, Deakin University.